Sorting by Reversals

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Outline

Motivation
- Symmetric group
- Reversal distance problem

MIN-SBR

Breakpoint graph

3/2-approximation
- Reversal graph
- Matching graph
- Approximation bound

Definition 1

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where $\pi_i = \pi(i)$, $\pi_0 = 0$, and $\pi_{n+1} = n + 1$.

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$id = (0\ 1\ \ldots\ n\ n+1) \in S_n$ is the *identity permutation*.

Definition 2

A reversal $\rho(i,j) \in S_n$ is defined as

$$\rho(i,j) = (0\ 1\ \cdots\ i-1\ j\ j-1 \cdots\ i+1\ i\ j+1\ \cdots\ n\ n+1)$$

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Example

Let $\pi = (0\ 1\ 3\ 4\ 2\ 5) \in S_4$.
Then

$$\pi \circ \rho(2,4) = (0\ 1\ 2\ 4\ 3\ 5).$$

Definition 3 (reversal distance problem)

Given two permutations $\sigma, \tau \in S_n$ find a sequence of reversals $\rho_1, \ldots, \rho_d \in S_n$ such that

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Observation: The reversal distance between σ and τ is the same as the reversal distance between $\tau^{-1} \circ \sigma$ and id.

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$\implies d(\pi) \leq 2.$

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A pair of consecutive elements π_i and π_j is

- an adjacency if $\pi_i \sim \pi_j$; and
- a breakpoint if $\pi_i \not\sim \pi_j$.

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Corollary 6 (lower bound, Kececioglu et al.)

$d(\pi) \geq \left\lceil \frac{b(\pi)}{2} \right\rceil$ *for all* $\pi \in S_n$.

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Let $G(\pi) = (V, E)$ with

- vertices $V = [0, n+1]$ representing the elements of π; and
- edges $E = R \cup B$ with
  - a red edge for every breakpoint in π; and
  - a blue edge for every missing adjacency in π.

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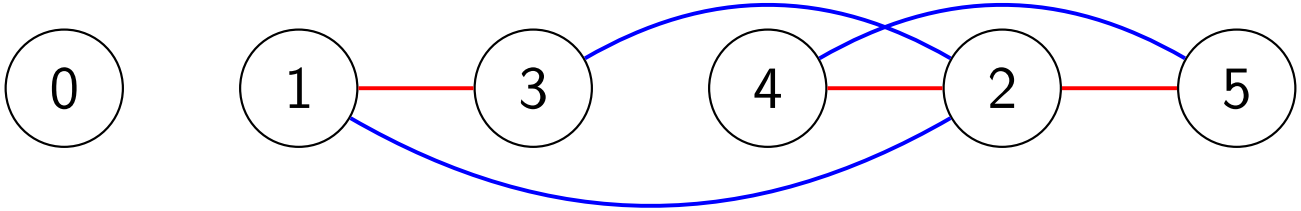

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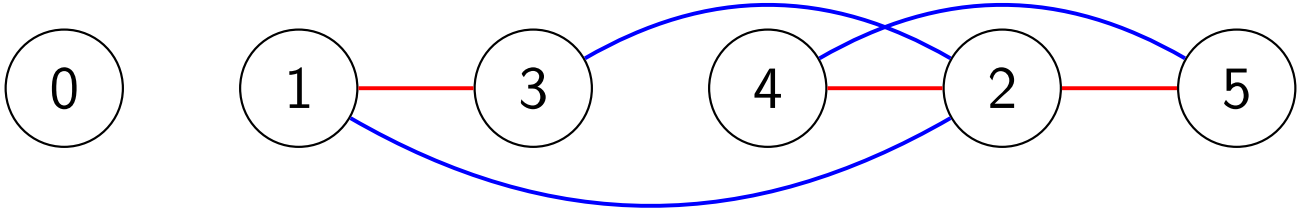

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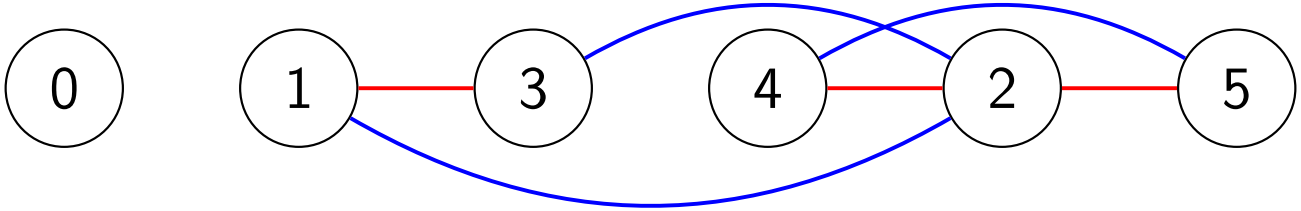

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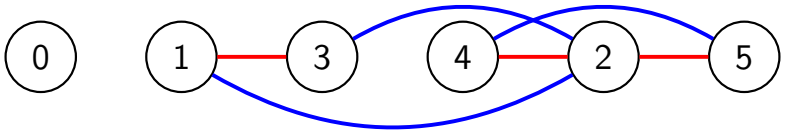
Corollary 8 (Bafna et al.)

$G(\pi)$ can be decomposed into edge-disjoint alternating cycles.



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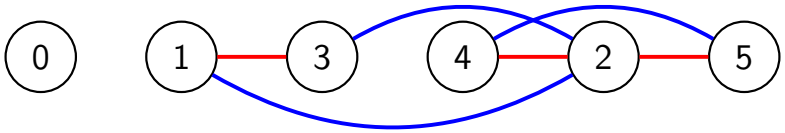
A reversal is called *k*-reversal if it removes k breakpoints.



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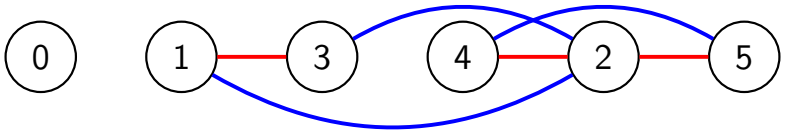
A reversal acts on two red edges of $G(\pi)$ if those two edges represent the breakpoints that are split apart by the reversal.



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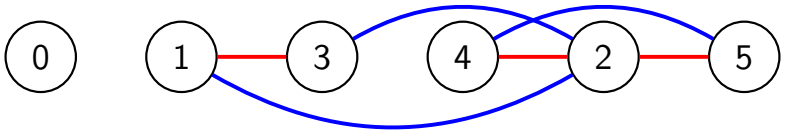

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We call an alternating cycle C in $G(\pi)$ oriented if there is a 1- or 2-reversal acting on two constituting red edges of C.

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Let $\pi, \rho \in S_n$ and ρ be a reversal. Then

$$b(\pi) - b(\pi \circ \rho) + c(\pi \circ \rho) - c(\pi) \leq 1.$$

Proof.

To show: $b(\pi) - b(\pi \circ \rho) + c(\pi \circ \rho) - c(\pi) \leq 1$.
We consider each case $b(\pi) - b(\pi \circ \rho) \in [-2, 2]$ separately.

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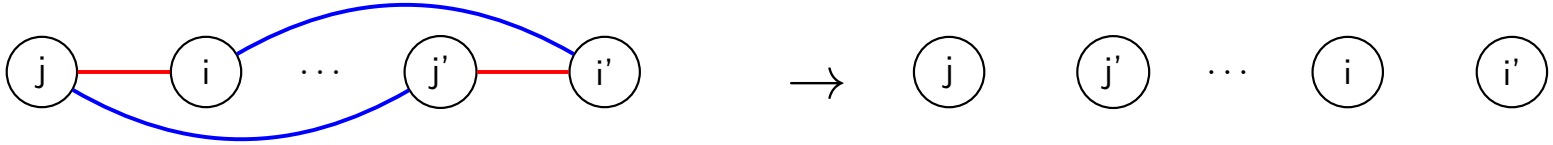

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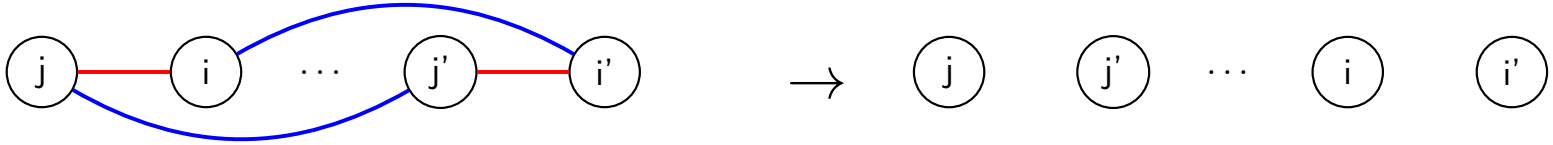

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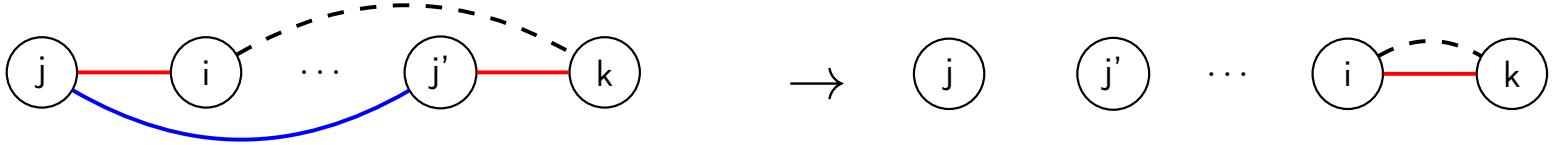

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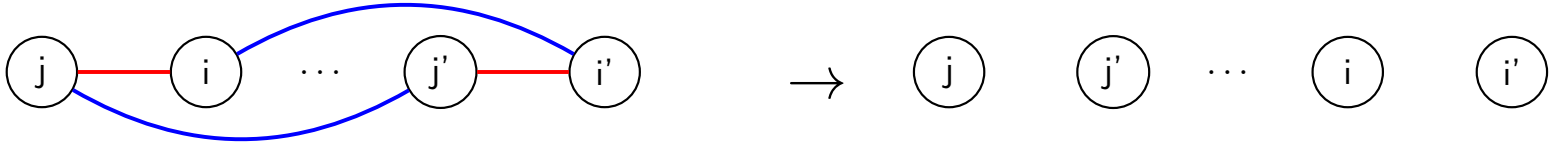

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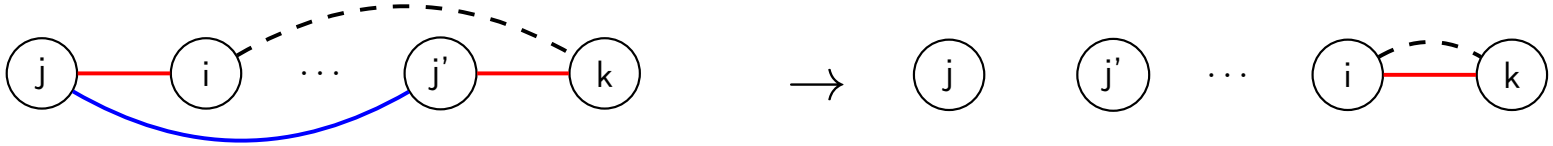

Proof for other cases similar. □

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Let $\pi \in S_n$. *Then*

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Setting $i = t$, proves the theorem. □

Theorem 12 (lower bound with 2-cycles, Christie)

Let $\pi \in S_n$ and $\mathcal{C}$ be a maximum alternating cycle decomposition of $G(\pi)$. Let $c_2(\pi)$ be the minimum number of alternating 2-cycles in any such $\mathcal{C}$. Then

$$d(\pi) \geq \frac{2}{3} b(\pi) - \frac{1}{3} c_2(\pi).$$

$\frac{3}{2}$-approximation

By theorem 12, an algorithm that finds a sorting sequence of reversals of at most length $b(\pi) - \frac{1}{2}c_2(\pi)$ achieves an approximation bound of $\frac{3}{2}$.

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Lastly, we prove the approximation bound.

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Observation: The only alternating cycle decomposition of $G(id)$ is $\mathcal{C} = \emptyset$.

Definition 13 (reversal graph, Christie)

Given an alternating cycle decomposition $\mathcal{C}$ of $G(\pi)$,
define $R(\mathcal{C})$ with

- an isolated blue vertex for each adjacency in π;
- m vertices for each m-cycle in $\mathcal{C}$ each representing the reversal $\rho(u)$ acting on the two red edges connected by a blue edge;
  - a vertex is red if the represented reversal is a 1- or 2-reversal
  - a vertex is blue otherwise
- connect two vertices with an edge if their corresponding blue edges *interleave*.

Observation: The only alternating cycle decomposition of $G(id)$ is $\mathcal{C} = \emptyset$.

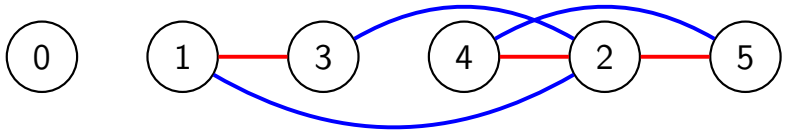
Corollary 14 (Christie)

$R(\emptyset)$ consists of $n+1$ isolated blue vertices.

Example

Let $\pi = (0\ 1\ 3\ 4\ 2\ 5) \in S_4$.
Given the alternating cycle decomposition $\mathcal{C}$ of $G(\pi)$

$$\mathcal{C} = \{(\{1,3\},\{2,3\},\{2,4\},\\ \{4,5\},\{2,5\},\{1,2\})\}$$



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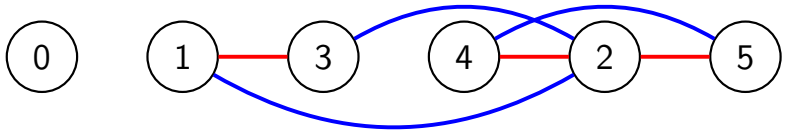
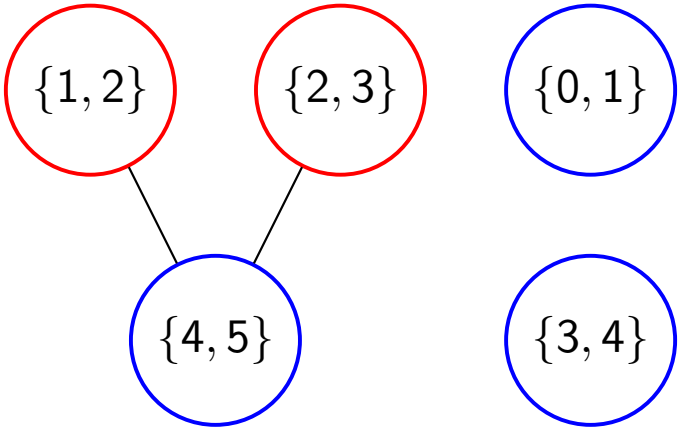


construct $R(\mathcal{C})$.



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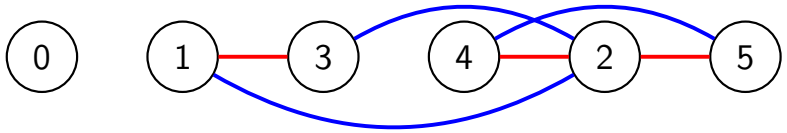

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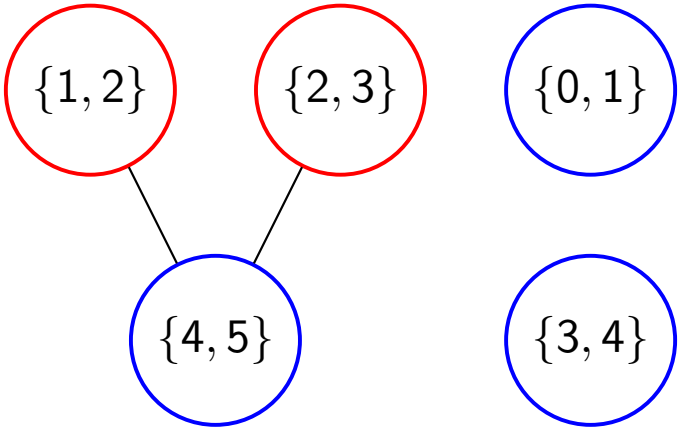

Idea: Each connected component of $R(\mathcal{C})$ can be sorted separately.

Let u be a vertex of $R(\mathcal{C})$ representing reversal $\rho(u)$.

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Definition 15

Denote by $\mathcal{C}_u$ the alternating cycle decomposition of $G(\pi \circ \rho(u))$ that is obtained from $\mathcal{C}$.

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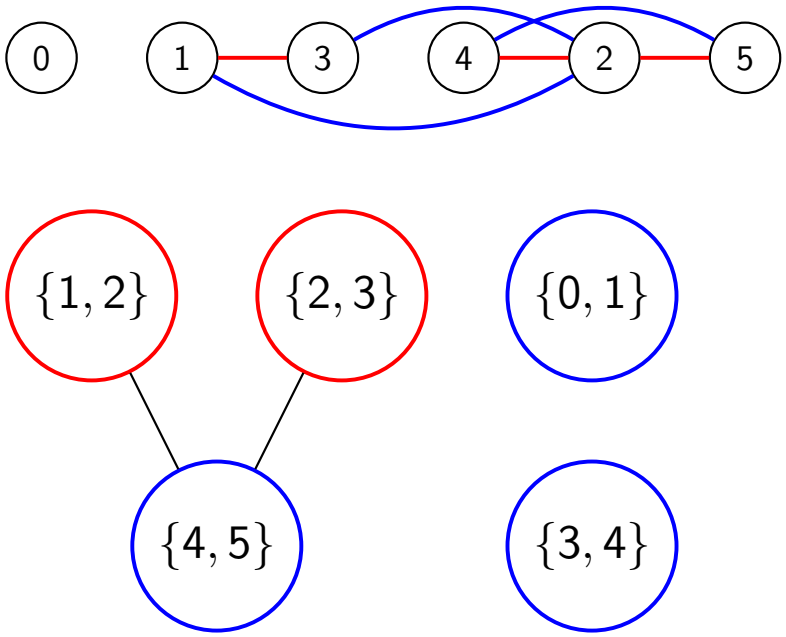
2. *flip the adjacency of every pair of vertices adjacent to u; and*
3. *if u is a red vertex, turn it into an isolated blue vertex.*

Example

Let $\pi = (0\ 1\ 3\ 4\ 2\ 5) \in S_4$ and $u = \{1, 2\}$.

$G(\pi)$ and $R(\mathcal{C})$	$G(\pi \circ \rho(u))$ and $R(\mathcal{C}_u)$
	

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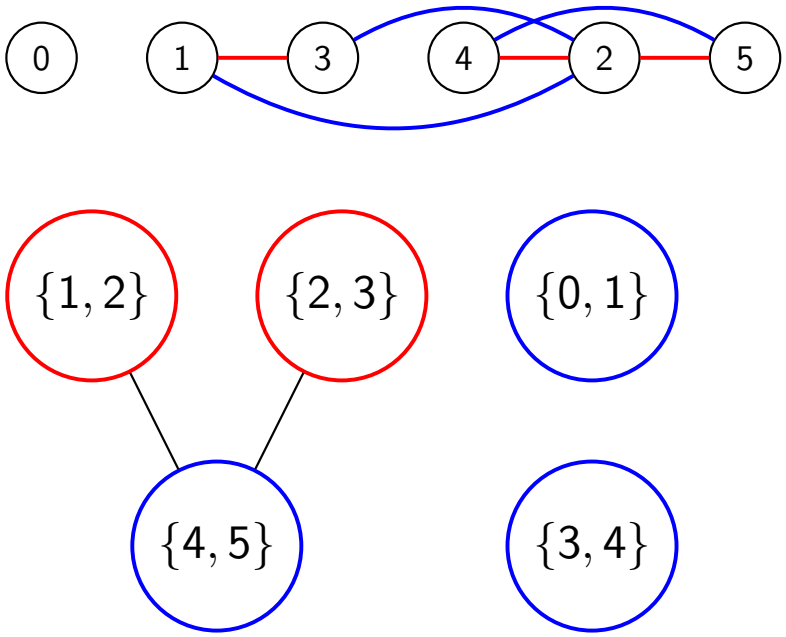	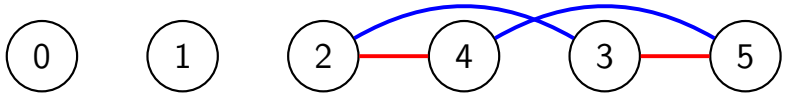

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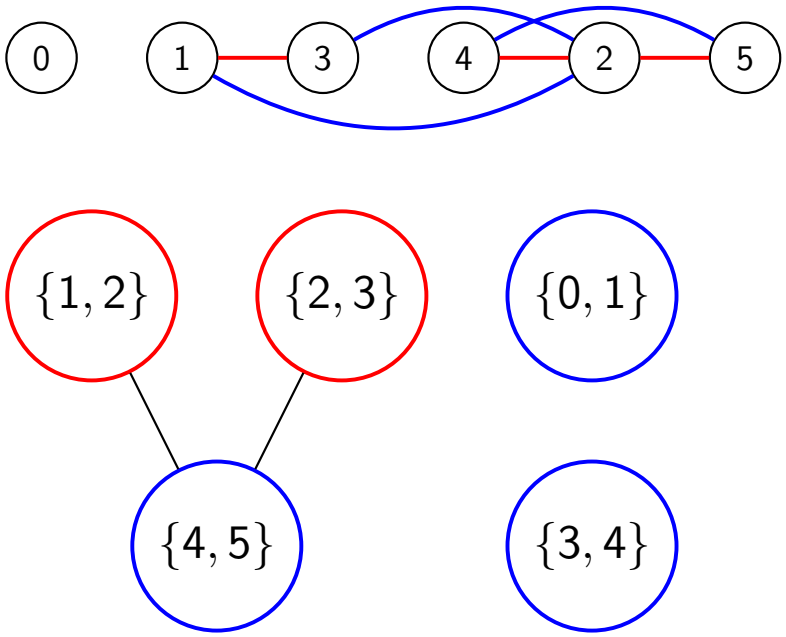
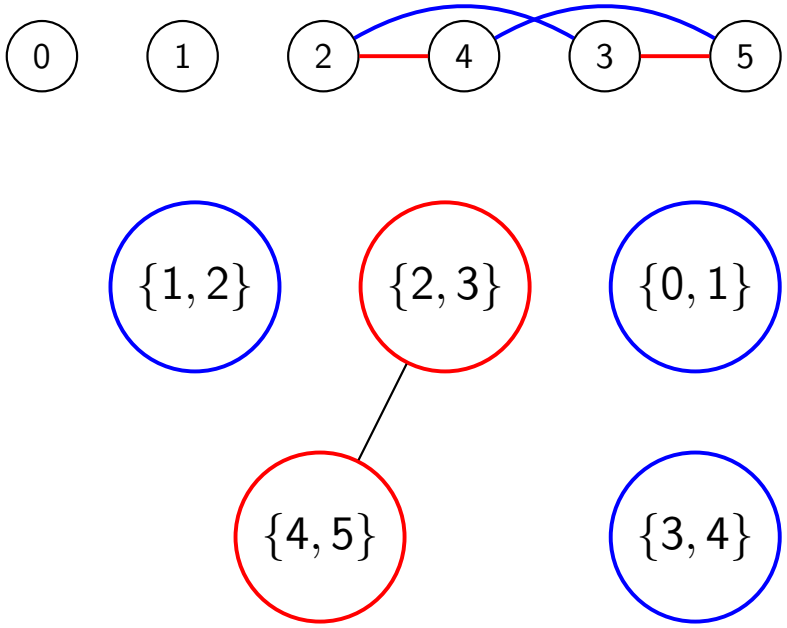
$G(\pi)$ and $R(\mathcal{C})$



$G(\pi \circ \rho(u))$ and $R(\mathcal{C}_u)$



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All vertices arising from the same alternating cycle in $\mathcal{C}$ are in the same connected component of $R(\mathcal{C})$.

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Lemma 18 (Christie)

Vertices arising from an unoriented 2-cycle of $\mathcal{C}$ must be in a connected component of $R(\mathcal{C})$ with vertices arising from at least one more alternating cycle of $\mathcal{C}$.

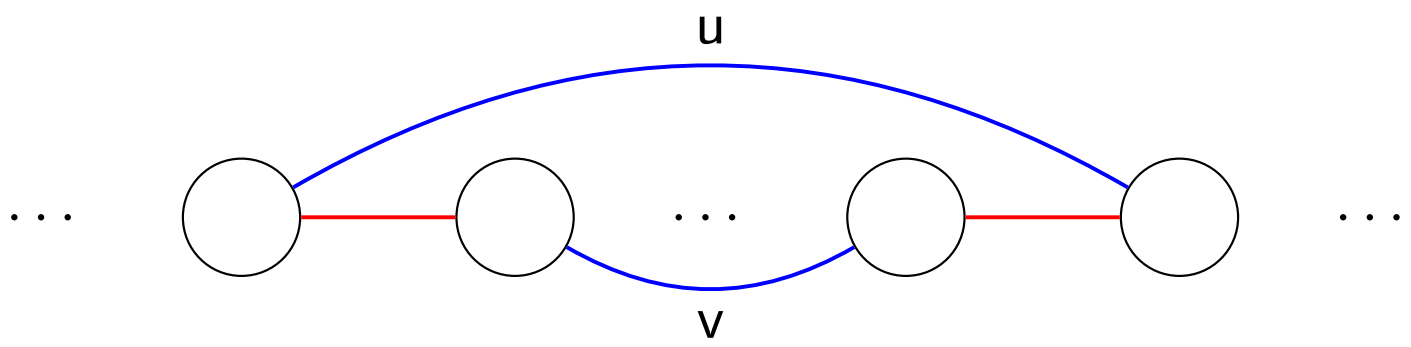

Figure 1: Unoriented 2-cycle

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We call a connected component of $R(\mathcal{C})$ oriented if it contains a red vertex or if it consists solely of an isolated blue vertex.

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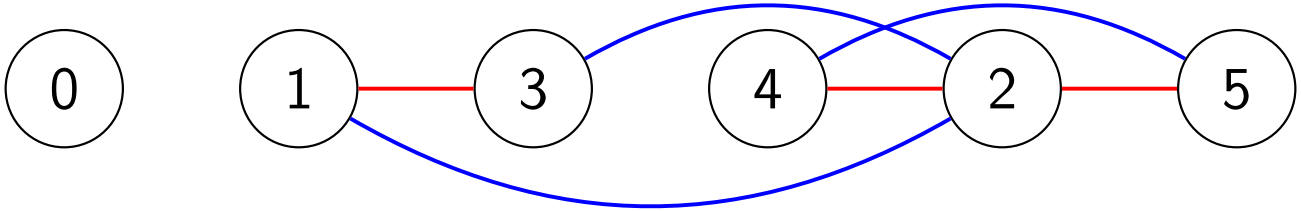
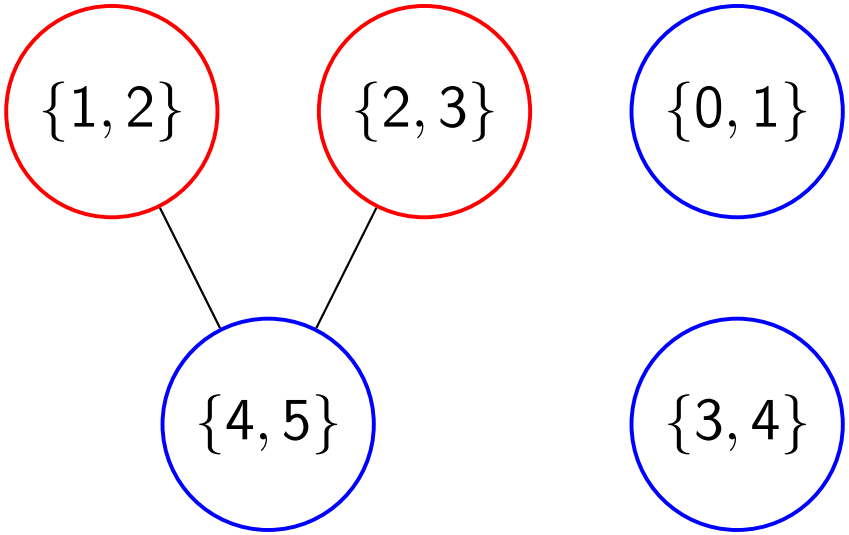
Lemma 20 (Christie)

If a connected component A of $R(\mathcal{C})$ is oriented and not an isolated blue vertex, it contains a red vertex u such that A_u is still oriented.

Example (elimination sequence)

Let $\pi = (0\ 1\ 3\ 4\ 2\ 5) \in S_4$





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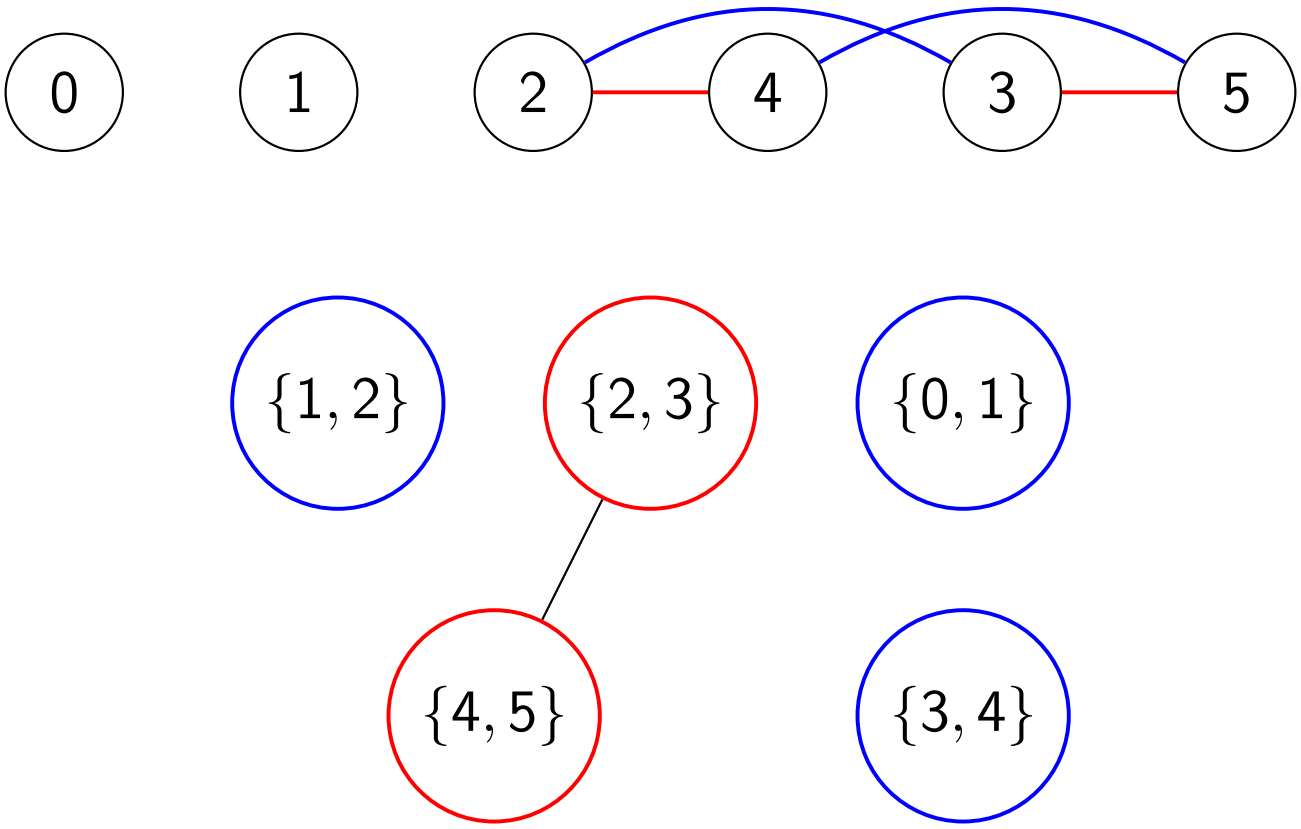

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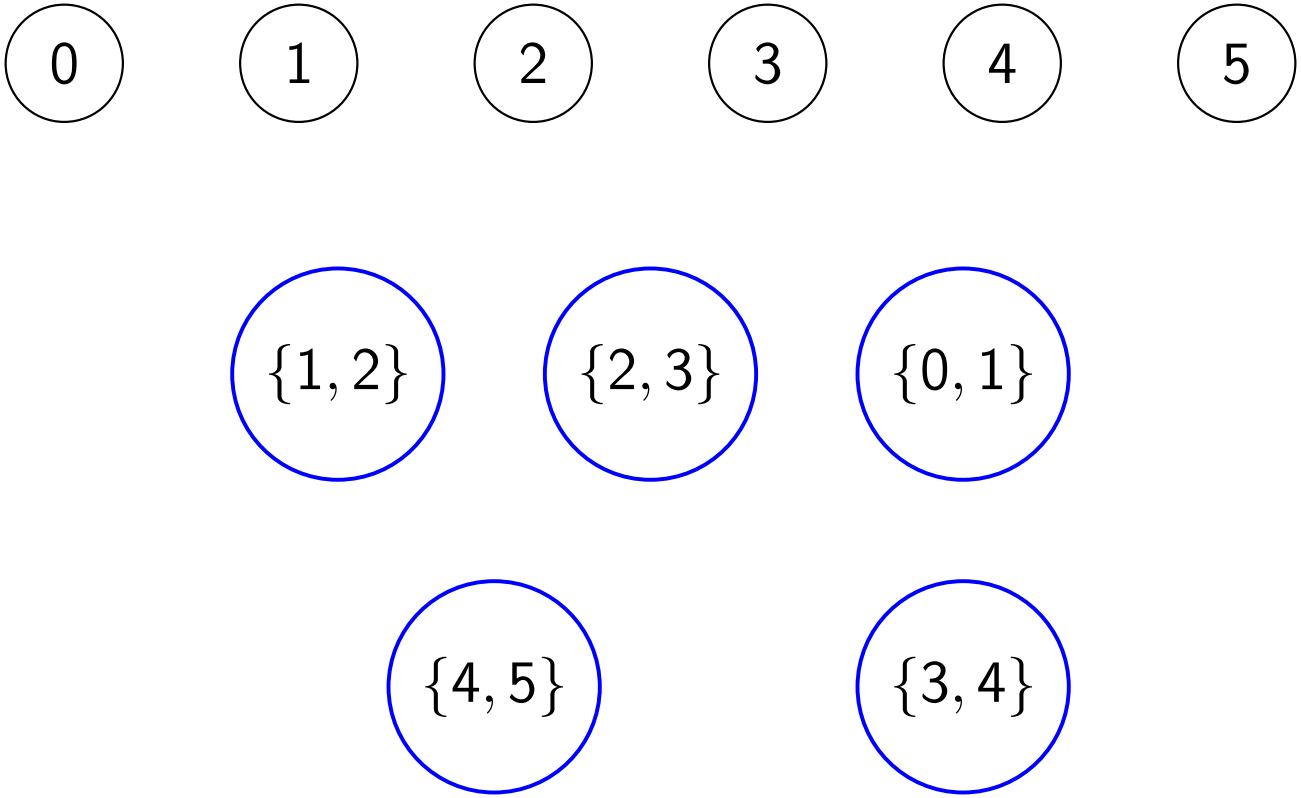

Observations:

- Every connected component A arising from k different alternating cycles of $G(\pi)$, eventually reduces to k 2-cycles.

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- An unoriented connected component requires one initial 0-reversal.

Theorem 21 (Christie)

Let $\pi \in S_n$ and $\mathcal{C}$ be an alternating cycle decomposition of $G(\pi)$. Then

$$d(\pi) \leq b(\pi) - |\mathcal{C}| + u(\mathcal{C})$$

where $u(\mathcal{C})$ is the number of unoriented components in $R(\mathcal{C})$.

Goal: Find a cycle decomposition of $G(\pi)$ that has a large number of 2-cycles.

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Idea 22

1. Construct a matching graph $F(\pi)$ where vertices represent red edges in $G(\pi)$ and vertices u, v are adjacent if they form a 2-cycle in $G(\pi)$.

Goal: Find a cycle decomposition of $G(\pi)$ that has a large number of 2-cycles.

Idea 22

1. Construct a matching graph $F(\pi)$ where vertices represent red edges in $G(\pi)$ and vertices u, v are adjacent if they form a 2-cycle in $G(\pi)$.
2. Find maximum cardinality matching M of $F(\pi)$.

Goal: Find a cycle decomposition of $G(\pi)$ that has a large number of 2-cycles.

Idea 22

1. Construct a matching graph $F(\pi)$ where vertices represent red edges in $G(\pi)$ and vertices u, v are adjacent if they form a 2-cycle in $G(\pi)$.
2. Find maximum cardinality matching M of $F(\pi)$.
3. Use a ladder graph $L(M)$ with vertices representing 2-cycles in M and form connected components (*ladders*) with 2-cycles sharing a blue edge in $G(\pi)$.

Definition 23

We call a 2-cycle selected if its corresponding edge of $F(\pi)$ is in M.

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Theorem 24 (Christie)

Given a maximum cardinality matching M of $F(\pi)$ it is possible to find an alternating cycle decomposition $\mathcal{C}$ of $G(\pi)$ that contains at least $\left\lceil \frac{y}{2} \right\rceil$ ladder 2-cycles and z independent 2-cycles.

Theorem 25 (Christie)

Let $\pi \in S_n$. Then

$$d(\pi) \leq b(\pi) - \frac{1}{2} c_2(\pi).$$

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$$d(\pi) \leq b(\pi) - \frac{1}{2}c_2(\pi).$$

Proof.

Using theorem 24, first find an alternating cycle decomposition $\mathcal{C}$ of $G(\pi)$ with at least $\left\lceil \frac{y}{2} \right\rceil$ 2-cycles as part of ladders and z independent 2-cycles.

Proof (cont.)

- Let k be the number of 2-cycles in oriented connected components of $R(\mathcal{C})$.

Proof (cont.)

- Let k be the number of 2-cycles in oriented connected components of $R(\mathcal{C})$.
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Proof (cont.)

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- Let v be the number of remaining unoriented connected components consisting only of vertices representing m independent selected 2-cycles.

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By theorem 21, we can sort π using at least $k + l + u + m$ 2-reversals and only $u + v$ 0-reversals.

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Left to show: $-k - l - m + v \leq -\frac{1}{2}c_2(\pi)$.

Proof (cont.)

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Proof (cont.)

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1. $k + l + m \geq \left\lceil \frac{y}{2} \right\rceil + z$ as $\left\lceil \frac{y}{2} \right\rceil + z$ is the number of selected 2-cycles in $\mathcal{C}$

Proof (cont.)

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3. $|M| = y + z \geq c_2(\pi)$.

Proof (cont.)

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1. $k + l + m \geq \lceil \frac{y}{2} \rceil + z$ as $\lceil \frac{y}{2} \rceil + z$ is the number of selected 2-cycles in $\mathcal{C}$;
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$$k + l + m - v \geq \left\lceil \frac{y}{2} \right\rceil + z - v \tag{1}$$

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$$k + l + m - v \geq \left\lceil \frac{y}{2} \right\rceil + z - v \tag{1}$$

$$\geq \left\lceil \frac{y}{2} \right\rceil + z - \left\lfloor \frac{z}{2} \right\rfloor \tag{2}$$

Proof (cont.)

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$$
\begin{aligned}
k + l + m - v &\geq \left\lceil \frac{y}{2} \right\rceil + z - v && (1) \\
&\geq \left\lceil \frac{y}{2} \right\rceil + z - \left\lfloor \frac{z}{2} \right\rfloor && (2) \\
&= \left\lceil \frac{y}{2} \right\rceil + \left\lceil \frac{z}{2} \right\rceil
\end{aligned}
$$

Proof (cont.)

Left to show: $k + l + m - v \geq \frac{1}{2} c_2(\pi)$. We know that

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&\geq \frac{1}{2} c_2(\pi) && (3)
\end{aligned}
$$

$\square$

Run time: $O(n^4)$, can be improved to $O(n^2)$ (Kaplan et al.).

Summary

- the number of alternating cycles in a breakpoint graph $G(\pi)$ is related to $d(\pi)$
- a sorting sequence of reversals can be constructed from an alternating cycle decomposition of $G(\pi)$

Outlook

- there exists a 1.375-approximation (Berman et al.)
- MIN-SBR for signed permutations is in P (Hannenhalli et al.)

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